

Introduction to Graphs.

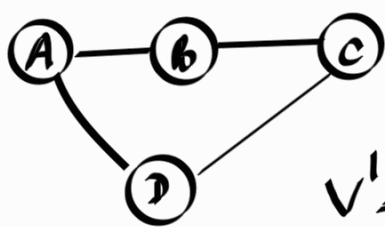
Def: Undirected Graph

An Undirected graph is a pair $G = (V, E)$ where V is a nonempty set of vertices or nodes, and $E \subseteq \{\{u, v\} : u, v \in V\}$ is a set of edges joining pairs of vertices.

Edges represents connection between vertices.

$$\textcircled{A} \quad V = \{A\}$$
$$E = \{\} = \phi$$
$$G = (V, E)$$

$$\textcircled{A} \text{ --- } \textcircled{B} \quad V = \{A, B\}$$
$$E = \{\{A, B\}\}$$
$$E = \{\{B, A\}\}$$
$$G = (V, E)$$


$$V' = \{A, B, C, D\}$$
$$E' = \{\{A, B\}, \{B, C\}, \{C, D\}, \{D, A\}\}$$
$$G' = (V', E')$$

$$\textcircled{A} \quad \textcircled{B}$$
$$V = \{A, B\}$$
$$E = \{\} = \phi$$

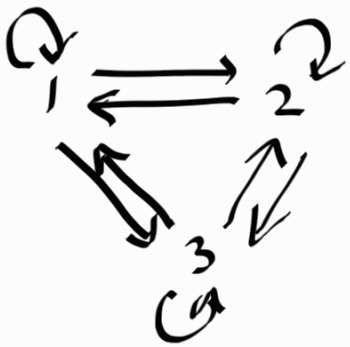
$\textcircled{A} \text{ --- }$ not a valid graph

All edges needs endpoints

Real-world example

- Facebook friendships
 - nodes can be people in FB
 - Edge between two people/nodes, if these two people are friends on FB
- Blood relationships

Q: what properties would mathematical relations need to have to be represented by an undirected graph?



If your relation is symmetric you can use an undirected graph to represent the mathematical relation.

Def: A graph is called simple if it contains no parallel edges or self-loops.



$$V = \{A, B\}$$

$$E = \{ \{A, B\}, \{A, B\} \}$$



Does not allow this



can not have these self loops in a simple graph.

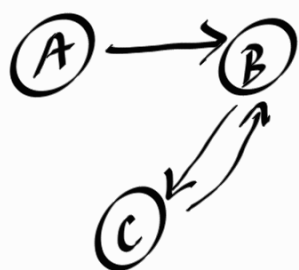
If a graph contains any of the above it is not considered as a simple graph.

$(A) \rightleftarrows (B) \leftarrow$ This is considered as simple

Def Directed graphs

A directed graph $G = (V, E)$, where V is a nonempty set of nodes, and

$E \subseteq V \times V$ is a set of edges joining (ordered) pairs of vertices.



$$V = \{A, B, C\}$$

$$E = \{(A, B), (C, B), (C, A)\}$$



$$V = \{A\}$$

$$E = \emptyset$$



$$V = \{A, B\}$$

$$E = \{(A, B)\}$$

$\neq$



$$V = \{A, B\}$$

$$E = \{(B, A)\}$$

real-world example

- Twitter follower relations

- node can be used to represent users,

- Edge from vertex a to b to represent a follows b .



	self-loops	parallel edges
Social Network	No	No
The web	Yes	Yes
flight-network	No	yes
Email network	yes	No

Let $e = \{u, v\}$ or (u, v)

- nodes u, v are adjacent or neighbors.



- In a directed graph, v is called out-neighbor of u & u is called an in-neighbor of v .



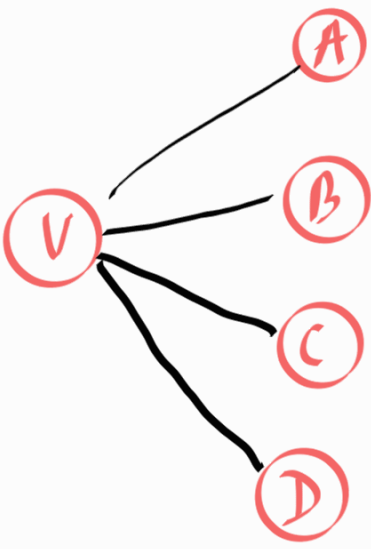
- u, v are endpoints of e
- u, v are incident to e .

Let v be a node in a simple undirected graph.

$\text{degree}(v) = \deg(v) = d(v) = \#$ of neighbors of v .

$$= \left| \{u \in V : \{u, v\} \in E\} \right|$$

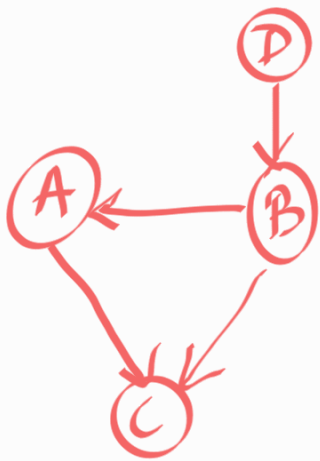
$$\deg(v) = 4$$



for directed graphs,

$\text{indeg}(v) = \#$ in-neighbors of v

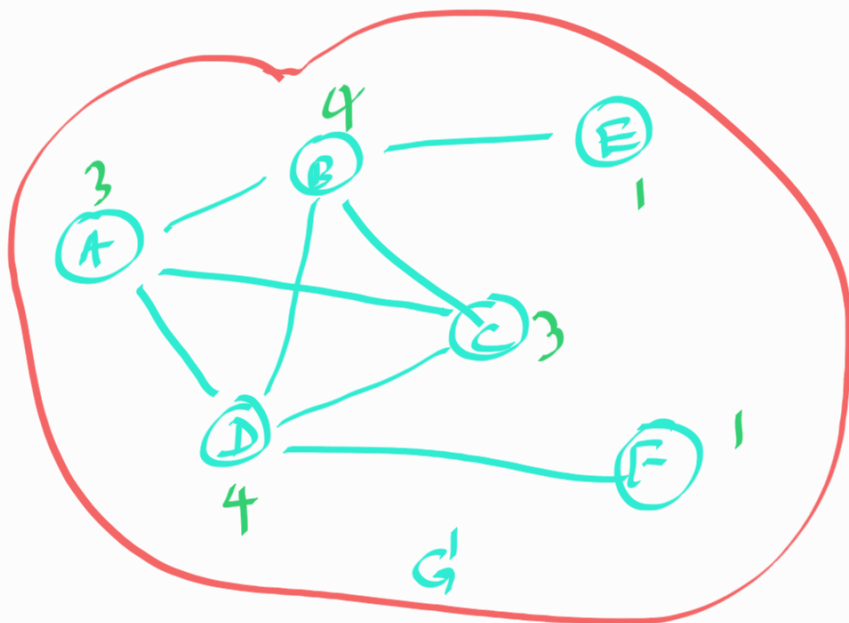
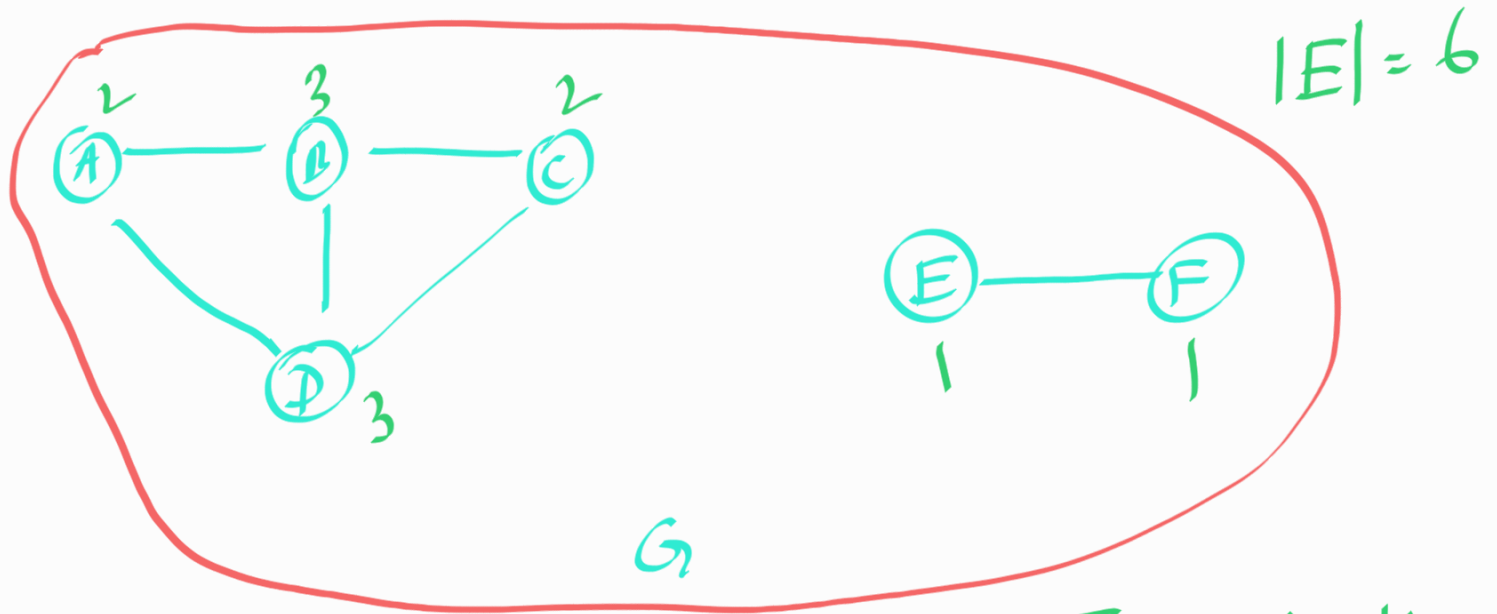
$\text{outdeg}(v) = \#$ out-neighbors of v



$$\text{indeg}(B) = 1$$

$$\text{outdeg}(B) = 2$$

For each of the graphs, label each node v with $\deg(v)$, and find $\sum_{v \in V} \deg(v)$



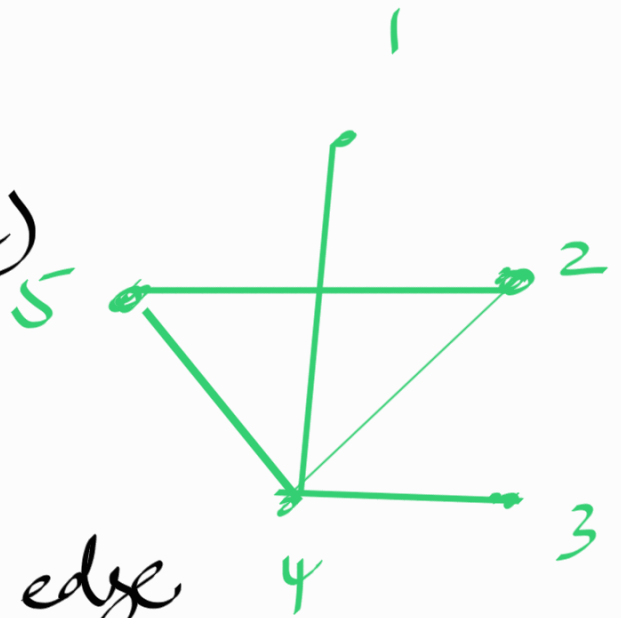
Theorem 11.8

Handshaking Lemma

Let $G = (V, E)$ be an undirected graph.

$$\sum_{v \in V} \deg(v) = 2 \cdot |E|$$

Proof: Let $G = (V, E)$ be an undirected graph.



Notice that every edge is connected to exactly two nodes, meaning that it contributes 1 to degree of two nodes that are connected by it.

$$\therefore \sum_{v \in V} \deg(v) = 2 \cdot |E|$$

Corollary $\rightarrow$ fact that follows
from a previous
lemma or theorem.

Corollary

let n_{odd} denote the # of
nodes whose degree is odd.

Then n_{odd} is even.

Corollary: Given $G=(V,E)$ a simple undirected graph, n_{odd} is even.

Assume the negation of the claim
 n_{odd} is odd.

$$\sum_{v \in V} \deg(v) = 2 \cdot |E| \leftarrow \text{even}$$

odd values

$$\sum_{v \in V} \deg(v) = \sum_{x \in \{y \in V : \deg(y) \text{ is odd}\}} \deg(x) +$$

even value $\rightarrow$

$$\sum_{x \in \{z \in V : \deg(z) \text{ is even}\}} \deg(x)$$

$$V = \{y \in V : \deg(y) \text{ is odd}\} \cup \{z \in V : \deg(z) \text{ is even}\}$$

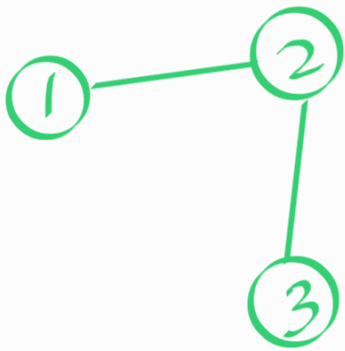
even = odd + even

but this is a contradiction

we cannot create an even number by summing odd number and an even number.

∴ the initial claim is true.

Def, A Graph G is called a complete graph or "Kleek" clique, if $G = (V, E)$ s.t. $\forall u, v \in V: u \neq v \Rightarrow \{u, v\} \in E$



is this a clique?

No, because

$\exists u, v$ s.t. $\{u, v\} \notin E$

let $u = 1$ $v = 3$

$\{1, 3\} \notin E$

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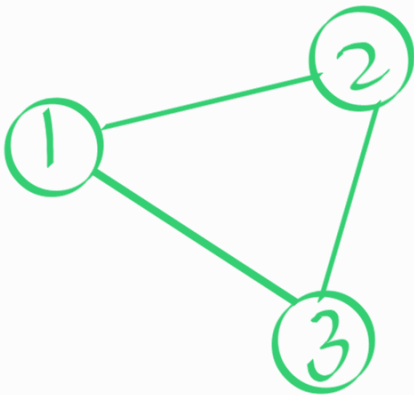
is this a clique?

Yes, because we cannot pick two distinct nodes, therefore the implication is vacuously true.

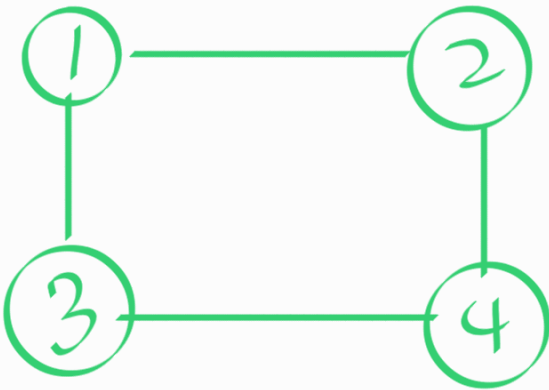


is this a clique?

yes

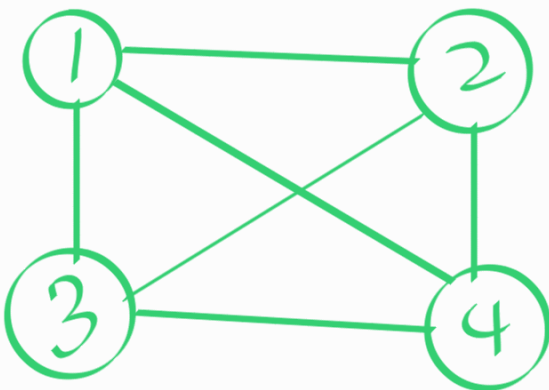


yes

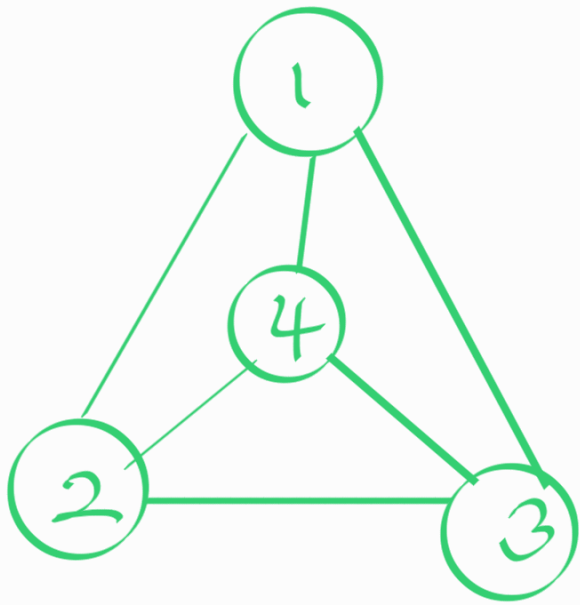


No because

$\{1, 4\} \notin E$



yes



yes.

cliques are so important in CS/Math we give a special notation to them.

we say:

A clique / complete graph on n nodes is denoted as K_n

K_1

$|V|=1$
 $|E|=0$



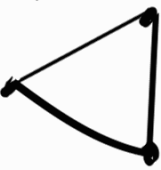
K_2

$|V|=2$
 $|E|=1$



K_3

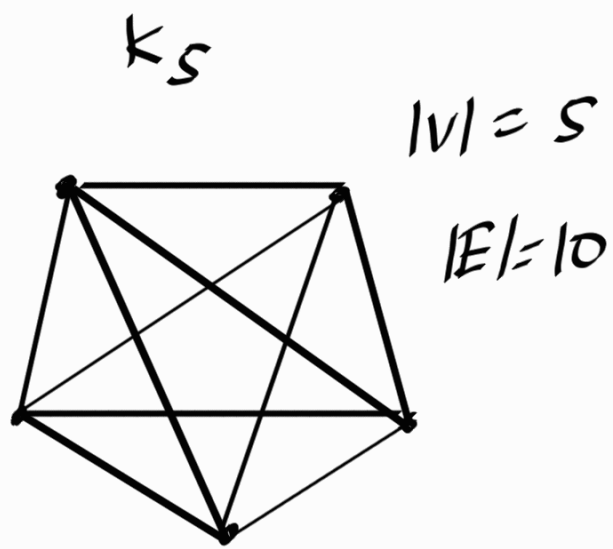
$|V|=3$
 $|E|=3$



K_4

$|V|=4$
 $|E|=6$





Claim: K_n has $\frac{n \cdot (n-1)}{2}$ edges.

we are going to prove this claim using different techniques

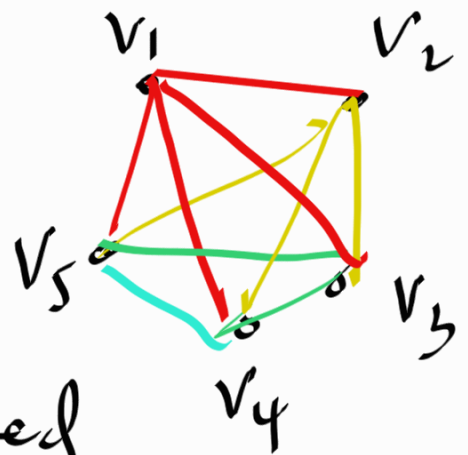
Proof #1: we give a way (algorithm) to count the # edges and show that it is

$$\frac{n \cdot (n-1)}{2}$$

Suppose we have a complete graph K_n . Label the vertices as $v_1, v_2, v_3, \dots, v_n$.

Starting with v_1 , count the uncounted edges adjacent to v_1 , and add them to the total.

v_1 has $n-1$ uncounted edges.



v_2 has $n-2$ uncounted edges

v_3 has $n-3$ uncounted edges

$\vdots$
 V_{n-1} has 1 uncounted edge

V_n has 0 " "

$$\begin{aligned} m = |E| &= 0 + 1 + 2 + 3 + \dots + (n-1) \\ &= \frac{n(n-1)}{2} \end{aligned}$$

Proof #2: using handshake lemma.

Note that every node in K_n has $n-1$ degree.

Since, K_n is a simple undirected graph we can use handshake lemma

$$\sum_{v \in V} \deg(v) = 2 \cdot |E|$$

$$\sum_{v \in V} (n-1) = 2 \cdot |E|$$

$$(n-1) \cdot n = 2 |E|$$

$$|E| = \frac{n(n-1)}{2}$$



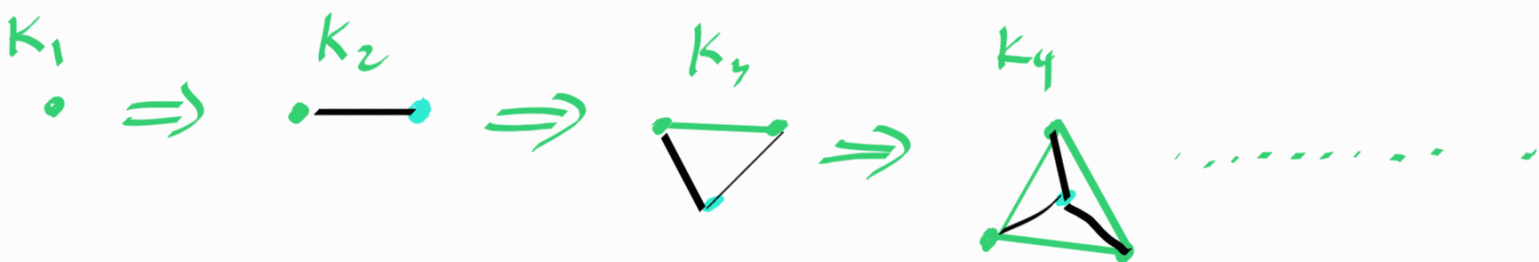
Proof #3: Using induction.

claim: K_n has $\frac{n(n-1)}{2}$ edges.

First, I am going to show that K_n is a recursively defined structure.

Base case: Let K_1 be a graph with 1 vertex and 0 edges.

Recursive case: Defining K_n using K_{n-1} graph. Take any K_{n-1} graph and add a new vertex V_n and connect V_n to each of the existing vertices in K_{n-1} .



Let $P(n)$ denote that K_n has $\frac{n(n-1)}{2}$ edges.

$$P(n) = \begin{cases} T, & \text{if } K_n \text{ has } \frac{n(n-1)}{2} \text{ edges} \\ F, & \text{otherwise} \end{cases}$$

We prove $\forall n \geq 1: P(n)$

Base case: Let K_1 be the graph with 1 vertex and 0 edges. We want to show that K_1 has $\frac{0 \cdot (-1)}{2} = 0$ edges.

This is obviously true.

Base case is proven.

Inductive case: $\forall n \geq 2: P(n-1) \Rightarrow P(n)$

Assume $P(n-1)$ (Inductive hypothesis)

K_{n-1} has $\frac{(n-1)(n-2)}{2}$ edges.

WTS: $P(n)$ is true, which is K_n has $\frac{n(n-1)}{2}$ edges.

Let K_n be a clique of n nodes.

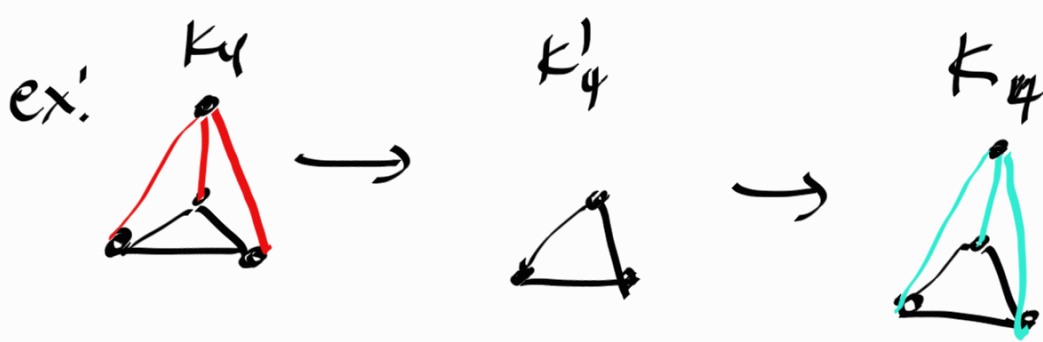
Let K'_n be the graph created by removing 1 node from K_n and all its incident edges.

Note that $K'_n = K_{n-1}$

using inductive hypothesis K'_n has $\frac{(n-1)(n-2)}{2}$ edges.

We add the node that we removed from K_n to create K'_n and all its previous edges connected to it. Then we would get K_n again.

Then # edges in K_n would be $\frac{(n-1)(n-2)}{2} + n-1$



$$\frac{(n-1)(n-2)}{2} + (n-1)$$

$$\frac{(n-1)}{2} [n-2 + 2] = \frac{n(n-1)}{2}$$

K_n has $\frac{n \cdot (n-1)}{2}$ edges

$\therefore P(n)$ is true,

Hence, using principle of mathematical induction $[\forall n \geq 1: P(n)]$

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